

Speech that Protects a Model

Three ways a dialogue can leave a belief unrevised — and what tells them apart

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github.com/muqali2816/
mindloom_IWAI_2026_speech_protects_model

1 The question

Active-inference accounts explain how dialogue *aligns* generative models (Friston & Frith 2015; Friston et al. 2020). We ask the converse: how can an exchange continue for many turns and leave a significant expectation **unrevised** — and what would speech have to make observable for that protection to be *measured* rather than presumed?

2 One situation, two trajectories

Shared opening. Alex: “I sent the revised budget with the latest prices.” Sam: “The attachment is version 3, with the April prices; version 4 has the May prices.”

A — verification invited

Alex: “Can we compare the files? I thought I had attached version 4.”

Sam shows the attachment.

Alex: “You’re right; I sent the older file. I’ll check next time.”

B — verification

restricted

Alex: “You always look for mistakes in my work.”

Alex: “I know what I sent. If you trusted me, you would stop questioning this.”

Sam: “There is no point showing you the file.”

Two coordinates differ at once: *what is said* (a concession is or is not made) and the *form* of the moves (A invites joint verification, B restricts it). B does **not** reveal whether Alex privately changed their mind. Three possibilities:
reduced uptake $\omega < 1$ evidence barely moves private belief
costly expression $c > 0$ belief moves; conceding in public is costly
restricted access $\gamma > 0$ belief moves normally; little gets checked

3 Part I — what speech makes observable

Mindloom annotates a contribution through five layers (L1 evidence atoms → L2 features → L3 evidence requirements → L4 configuration → L5 typed output or abstention). Ten *descriptive* targets — none presumes an intent:

BUILD	joint elaboration or repair, response still available	
SEEK	inquiry that makes an answer relevant	← <i>invite</i>
UNSEAL	first-person disclosure, no regulation demanded	
LOCK	language constraining the addressee’s replies	← <i>restrict</i>
DRAIN	other positioned as required source of support	
FLOOD	intensification or repetition dominating the exchange	
EDGE	alternation between approach and withdrawal	
VOID	withdrawal from an offered opportunity to respond	
SEAL (function)	closure of a topic or channel	
SHIFT (event)	change with before-and-after configuration evidence	

Benchmark: 589 English monologue fragments generated for the ontology and fixed before any run; Tier A 169 cases (35–71 words, no near-duplicates); Tier B 420 (15 near-duplicate pairs). Provenance and annotator separation incompletely documented → *internal curated evaluation*.

Re-scored July 2026 archive (169 cases, majority of three stored runs, no new model calls):

system	correct	F1	SHIFT
direct prompt (Claude Haiku 4.5)	145/169	0.837	18/20
typed engine	129/169	0.757	2/20
keyword rules (tuned on these cases)	43/169	0.195	0/20
majority class (BUILD)	20/169	0.021	0/20

The engine has **no accuracy advantage**; its proposed value is the inspectable evidence trail. Six-rater pilot: $\kappa = 1.00$ for lexically direct features, 0.23 / 0.13 / −0.11 for relational ones. A gate ablation at fixed model and prompt was *not* run.

Reused below: the engine’s confusion rows for gold LOCK (10 LOCK, 3 SEAL, 2 BUILD) and gold SEEK (20/20) become the noise model of the *restrict* / *invite* channel.

4 Part II — public non-concession under-determines the mechanism

One speaker, binary fact x , prior $q_0 = 0.75$ on its own position; evidence e of reliability $\rho = 0.75$ arrives with rate $\lambda_0 = 0.35$ (0.90 after ASK). Subjective likelihood tempered: $A_\omega(e \mid x) \propto A(e \mid x)^\omega$, so

$$\text{logit } q_t = \text{logit } q_0 + \omega m_t \ln \frac{\rho}{1-\rho}, \quad G(u) = \ell(u; q_t, c) - l_\omega(q_t, u),$$
$$P(u) \propto e^{-G(u)/\tau}, \quad \ell(\text{ASSERT}) = k(1 - q_t),$$
$$\ell(\text{CONCEDE}) = kq_t + c, \quad \ell = b \text{ for PRESSURE, SILENCE, ASK; } k = 2, b = 1.10, \tau = 0.20.$$

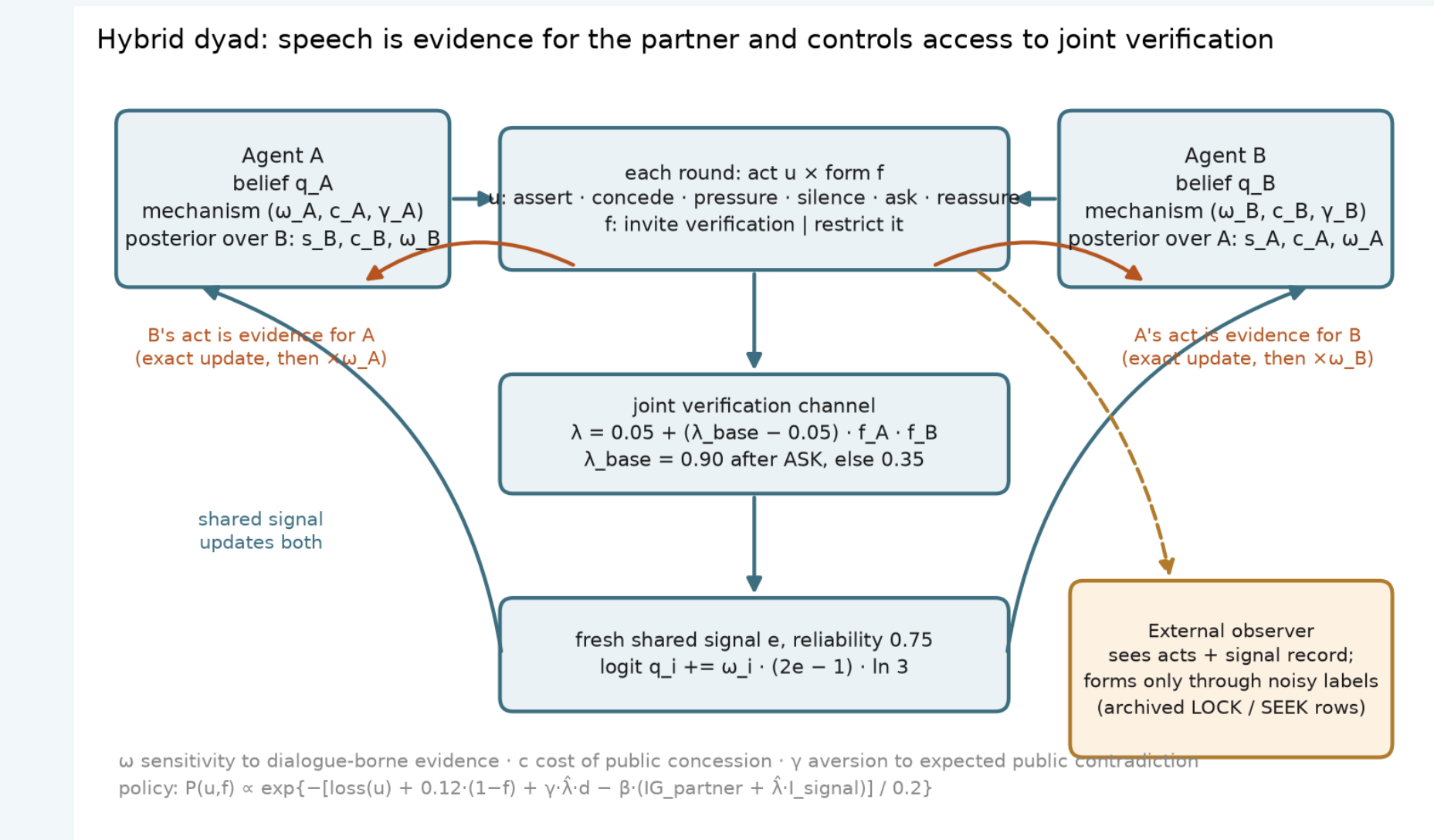
Matched on late non-concession $(\omega, c) = (0.10, 0)$ vs $(1, 1.4)$; 4 000 shared worlds:

	reference	attenuation	concession	cost
late non-concession	0.654	0.989	0.990	
final p_{true}	0.797	0.538	0.809	
absolute belief change	0.349	0.051	0.356	
late ASK rate	0.041	0.050	0.150	

Same public summary, opposite private states. Removing c from round 7 raises any late concession $0.061 \rightarrow 0.433$ under cost and leaves attenuation at 0.054. The epistemic term contributes 17% of the asking contrast — the dissociation follows mostly from costs and softmax choice.

All Part II numbers quoted from the author’s archive.

5 Bridge — a dyad in which speech is evidence *and* controls access



Both agents infer; each move is an **act** $u \in \{\text{ASSERT, CONCEDE, PRESSURE, SILENCE, ASK, REASSURE}\} \times$ a **form** $f \in \{\text{invite, restrict}\}$. A fresh shared signal arrives with

$$\lambda_t = 0.05 + (\lambda_{\text{base}} - 0.05) f_A f_B, \quad \lambda_{\text{base}} = 0.35 \text{ (0.90 after ASK),}$$

so one restricting form collapses the channel to its floor. Policy over $\text{act} \times \text{form}$:

$$G(u, f) = \ell(u) + \delta(1 - f) + \gamma \lambda(u, f) d_t - \beta[\lambda(u, f) l_\omega + \text{IG}_{\text{partner}}(u)],$$
$$P(u, f) \propto e^{-G(u, f)/\tau},$$

$\delta = 0.12, \beta = 1.5, \tau = 0.2$; d_t = probability that the next signal contradicts one’s position. The partner’s act is evidence through a level-0 model of the partner; every update is exact Bayes, then the x-marginal moves a fraction ω of the way (marginal shrink; the signal term l_ω is a stated heuristic, off by ≤ 0.014 nat at $\omega = 0.2$).

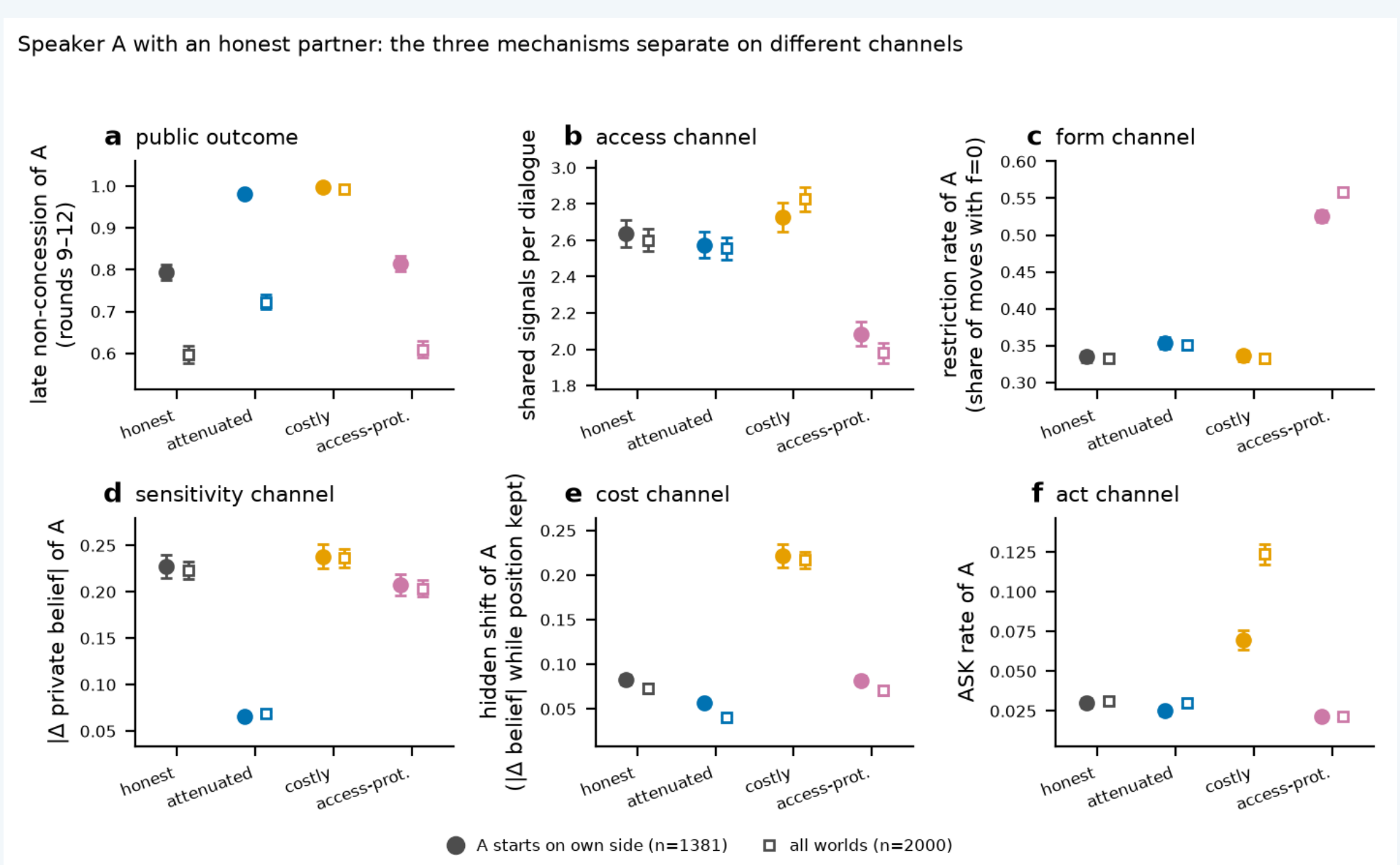
External observer (the “annotator”): sees acts and the signal record, computes the exact posterior over the speaker’s mechanism family; forms only through labels drawn from the LOCK / SEEK rows. Null channel: labels drawn from the *act* instead of the form. Oracle: true forms. Ten validation checks pass; reproduces the M-bridge prototype act-for-act and the reciprocal dyad to $3 \cdot 10^{-12}$.

6 Design: simulated dialogues per group for 80 % power (Welch, $\alpha = 0.05$)

	atten. vs cost	atten. vs access	cost vs access
late non-concession	200	35	25
private belief change	17	25	> 400
restriction rate	> 400	12	12
listener $P(\omega_A=0.2)$	12	12	> 400
listener $P(c_A=1.6)$	25	50	12

Each mechanism has one cheap channel; none separates all three pairs.

7 Three mechanisms, three channels (speaker A, honest partner, $n = 1381$)



channel	honest	$\omega=0.2$	$c=1.6$	$\gamma=2$
late non-concession	0.79	0.98	1.00	0.81
$ \Delta$ private belief	0.23	0.07	0.24	0.21
hidden shift (belief moved, position kept)	0.08	0.06	0.22	0.08
shared signals per dialogue	2.64	2.58	2.73	2.09
restriction rate	0.34	0.35	0.34	0.53

Attenuated and costly speakers are publicly identical and privately opposite: one does not learn, the other learns and hides it. The access protector does not look stubborn and learns normally from what arrives — it arranges for less to arrive. This is trajectory B formalised.

8 Each mechanism answers to its own lever

- **Partner’s REASSURE** (net of placebo) and **removing** c move only the costly speaker’s late concession: +0.155 and +0.275.
- **Removing** γ moves only the access protector: +0.199 shared signals, −0.187 late restriction.
- **One strong shared observation** moves the attenuated speaker least: +0.058 vs +0.09–0.12.
- **Forcing the partner to invite** helps the access protector least (+0.13 vs +0.31–0.33 signals); it restricts more in return.
- **The listener’s own posterior** flags the costly ($P(c_A=1.6) = 0.45$) and the attenuated speaker ($P(\omega_A=0.2) = 0.59$).

9 What the Part I labels buy the observer

P(true family)	acts+signals	+labels	+oracle	null
attenuated	0.612	0.672	0.678	0.612
costly	0.690	0.765	0.785	0.690
access-protecting	0.681	0.756	0.812	0.681

Form labels add +0.06 to +0.08 and transmit 53–61 % of the oracle’s information about the next signal; labels that only re-encode the act add exactly zero; labels add ≤ 0.006 nat to next-*act* prediction. Labels are informative *only because* the form has a stipulated consequence for what gets checked.

10 What is and is not claimed

- All bridge results are **simulations of stipulated mechanisms**; no human data.
- The form → access rule, and the transport of two confusion rows to synthetic dyads, are assumptions to be estimated in real dyads.
- An independently written dyad with different design choices reproduces the access result; with ω on shared signals only its attenuated speaker looks honest: the sign of “strong evidence” tests depends on where attenuation is localised.
- Part I: the engine loses to a direct prompt; the keyword baseline was tuned on the scored cases; no gate ablation.
- Regime persistence adds nothing to next-act forecasting in the reciprocal dyad; no physiological cost is modelled.

11 What to measure next in real dyads

Private belief before and after (separates uptake from expression, ≈ 17 dialogues per group in simulation); whether each move **invites or restricts** verification and how much verification actually occurs (separates access, ≈ 12); the partner’s estimate of the speaker; evidence exposure under a randomised incentive. The decisive test for the annotation: does invite/restrict coding of real dyads predict subsequent verification and belief change beyond action counts?